

PROJECT BUTTON HR–V0 ELECTRICAL V3 CONNECTED CANDIDATE

Separate RESET and ARM, two PNOZ stages, dual watchdog contacts, external adapters, redundant actuator interruption.

01 External listed sources and DC boundaries

File: 01\_external\_sources.kicad\_sch

02 Dual–channel E–stop and RESET eligibility

File: 02\_estop\_eligibility.kicad\_sch

03 Distinct ARM and watchdog eligibility

File: 03\_arm\_watchdog\_eligibility.kicad\_sch

04 Contactor coils, mirror contacts and EDM

File: 04\_contactor\_edm.kicad\_sch

05 Redundant actuator–power interruption

File: 05\_actuator\_interruption.kicad\_sch

06 Protected actuator branches and VDD–isolating injection

File: 06\_branches\_and\_injection.kicad\_sch

07 Independent watchdog power, controller and drivers

File: 07\_watchdog\_control.kicad\_sch

08 Calculated dual–channel 24 V watchdog feedback

File: 08\_watchdog\_feedback\_interface.kicad\_sch

09 Compute, debug and control terminals

File: 09\_compute\_and\_control\_terminals.kicad\_sch

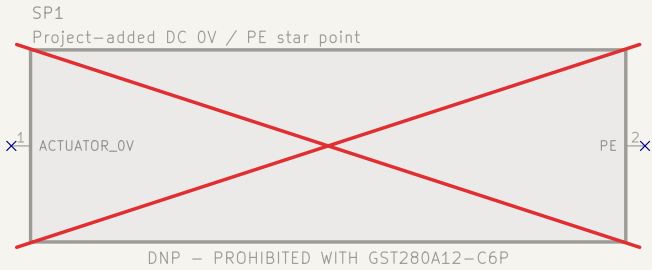
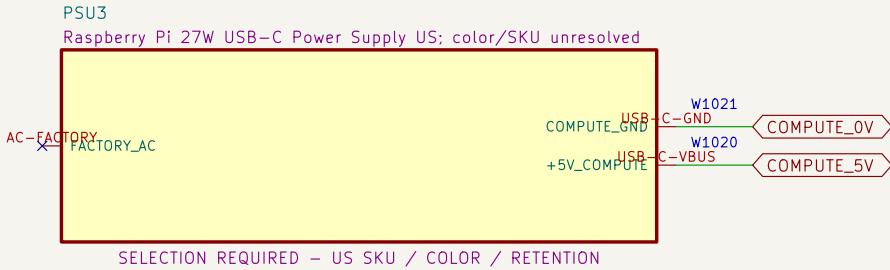
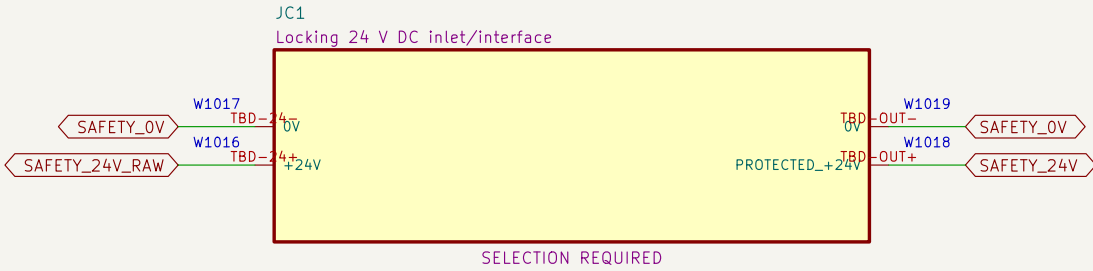
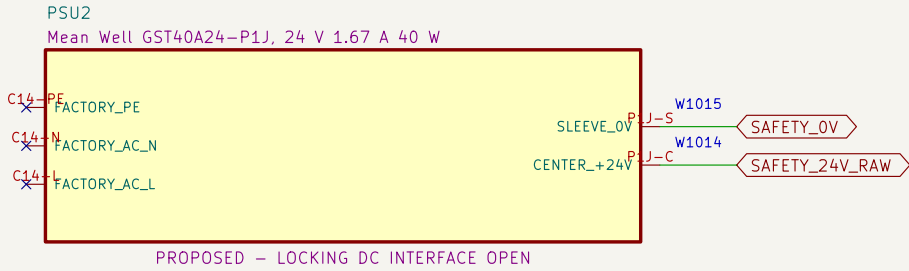
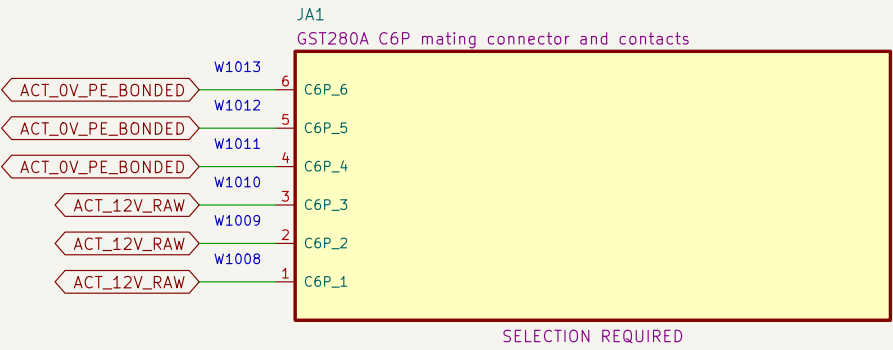
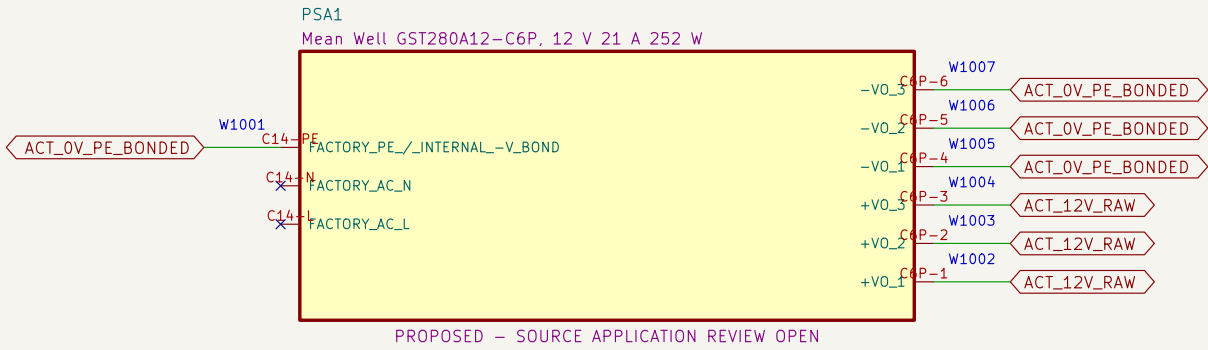
10 U2D2, actuator ports and bonding boundary

File: 10\_actuator\_interfaces.kicad\_sch

|                                                                                                                                    |                  |              |
|------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------|
| V2.1 PRESERVED; V3 IS A CONNECTED CANDIDATE<br>PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION<br><b>Project Button</b> |                  |              |
| Sheet: /<br>File: project–button–v3.kicad_sch                                                                                      |                  |              |
| <b>Title: Project Button HR–V0 Electrical V3 index</b>                                                                             |                  |              |
| Size: A3                                                                                                                           | Date: 2026–08–06 | Rev: V3–P0.6 |
| KiCad E.D.A. 10.0.5                                                                                                                |                  | Id: 1/11     |

01 External listed sources and DC boundaries

Factory-sealed adapters eliminate project-built mains wiring.

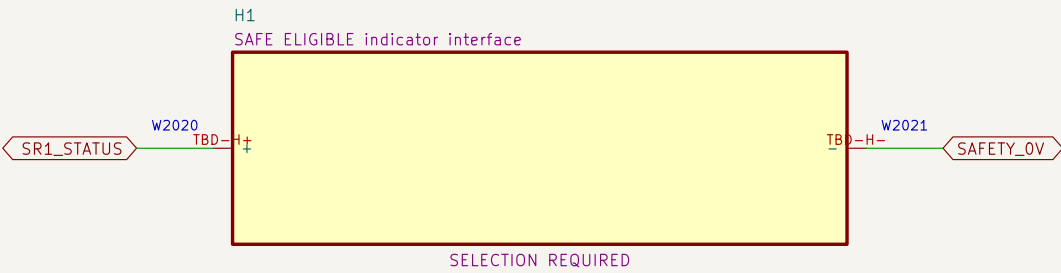
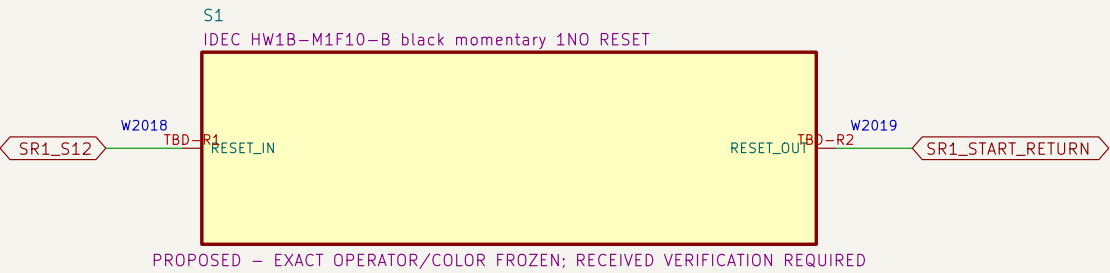
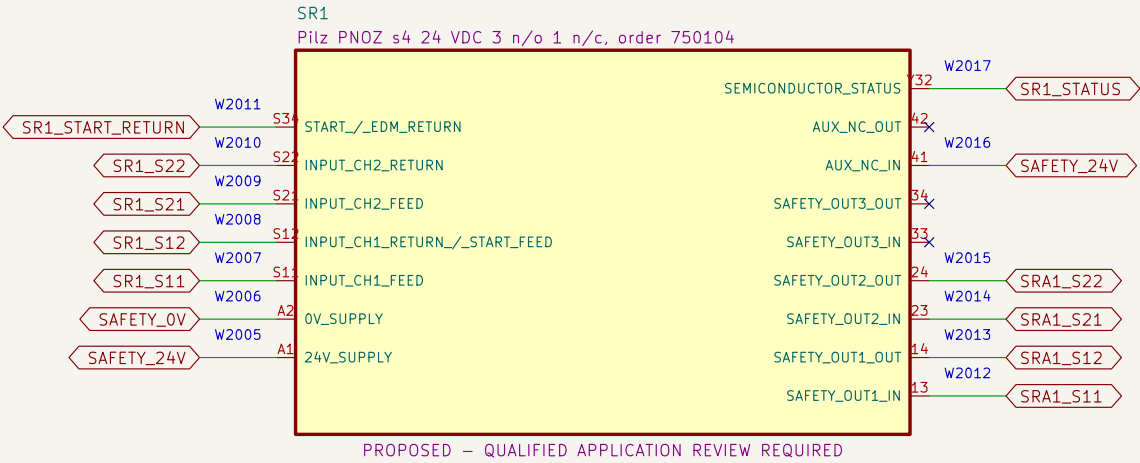
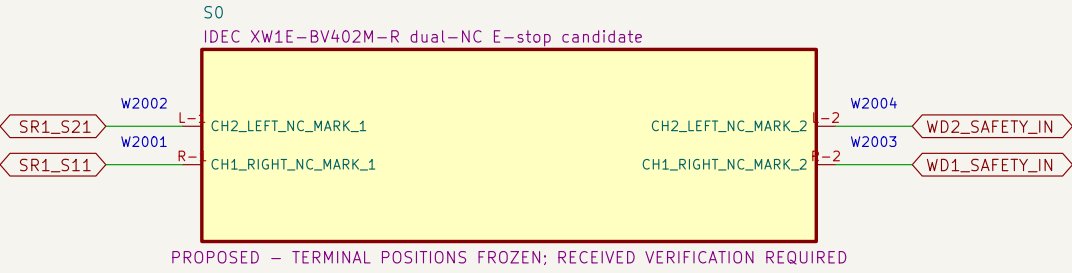


NOTE 1: All AC conductors shown are factory boundaries, not project-built wiring.

NOTE 2: Site cords, receptacles, GFCI/code basis and source application review remain open.

02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains one E-stop NC and one watchdog NO contact; RESET cannot energize K1/K2.

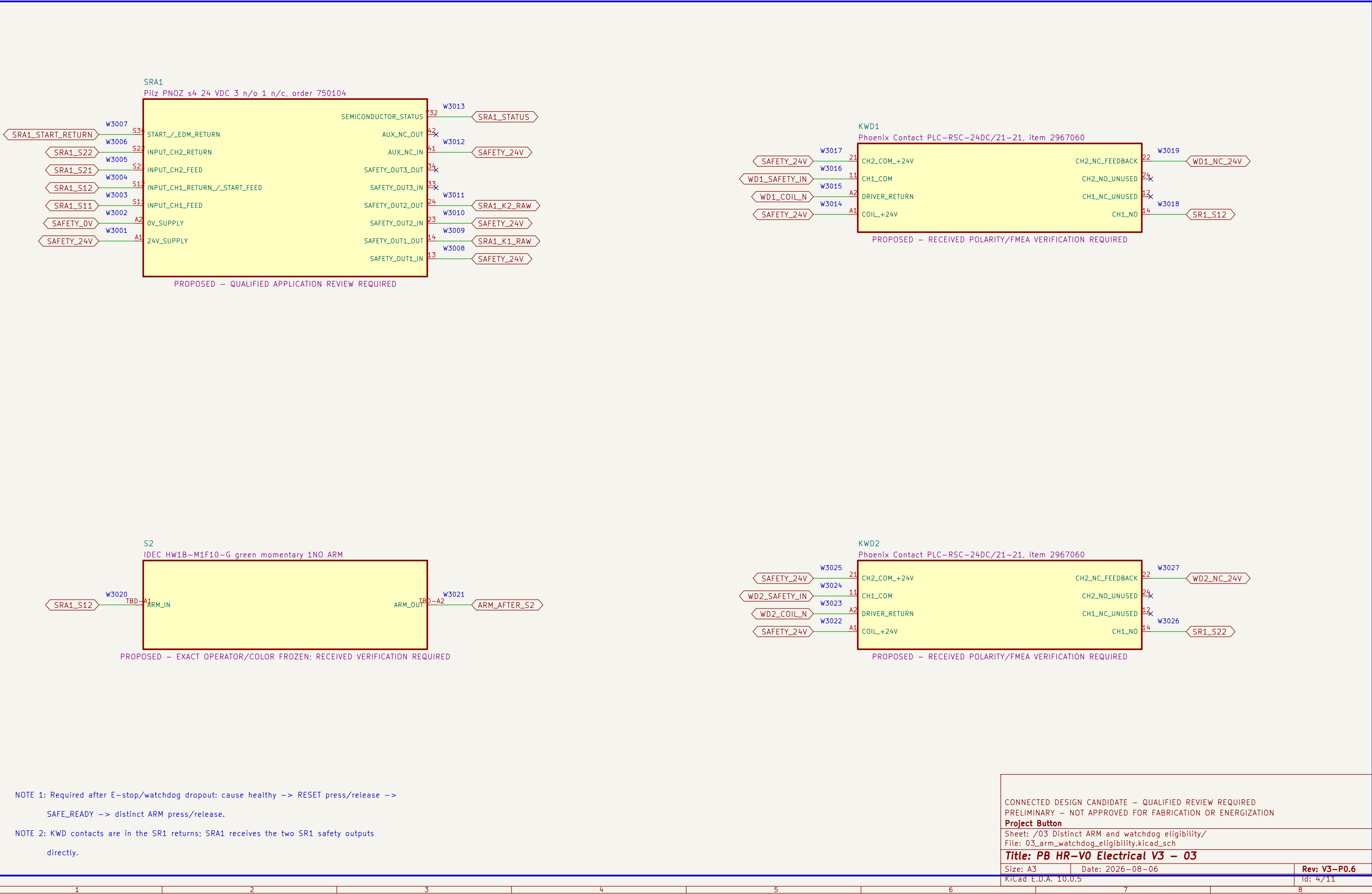


NOTE 1: Heartbeat loss opens both SR1 input returns; recovery alone cannot restore the monitored RESET stage.

NOTE 2: RESET release may make SR1 eligible, but SRA1 and K1/K2 remain de-energized until a later ARM.

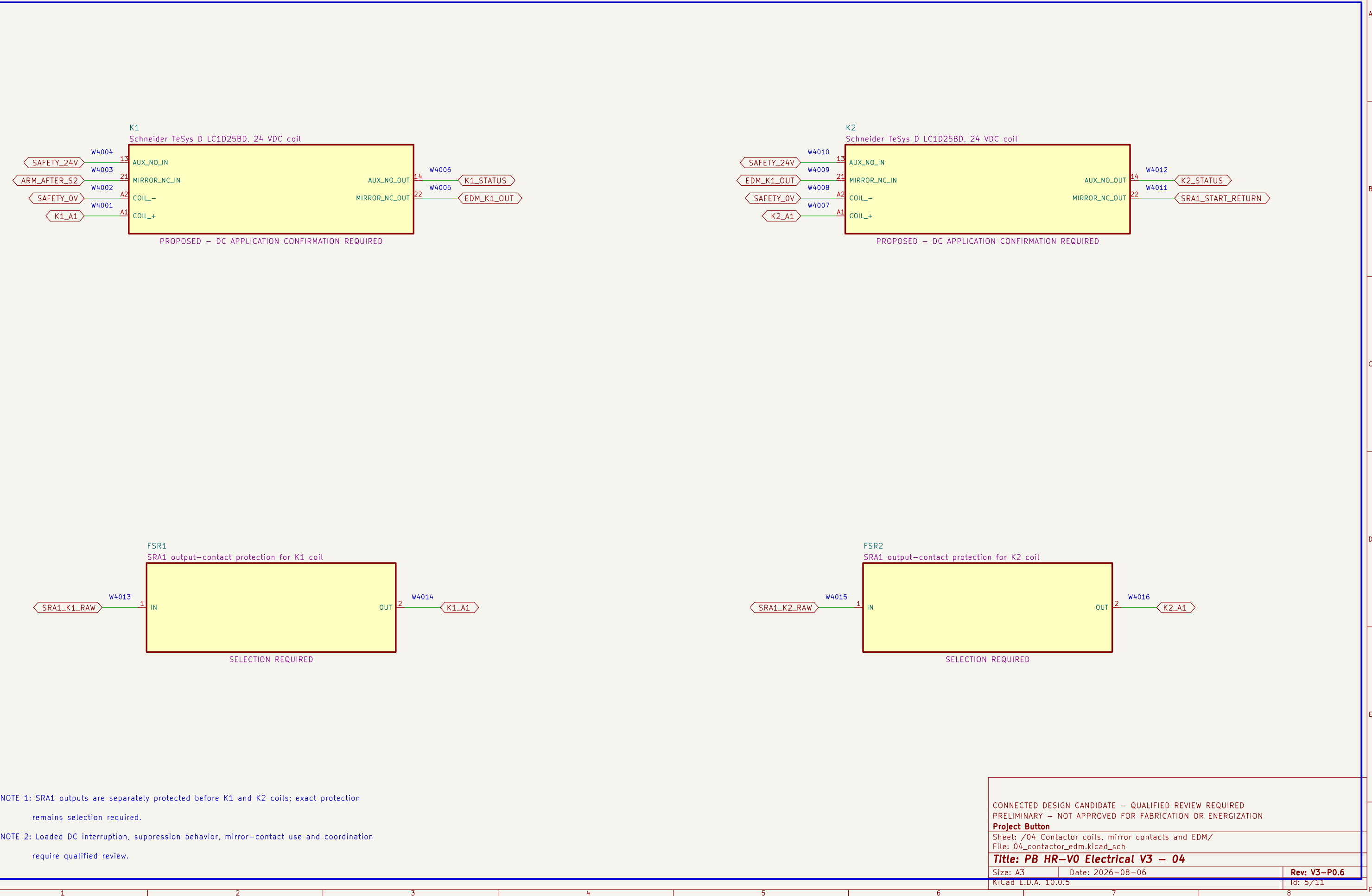
03 Distinct ARM and watchdog eligibility

SRA1 requires SR1 eligibility, two watchdog channels, EDM proof and a new ARM action.



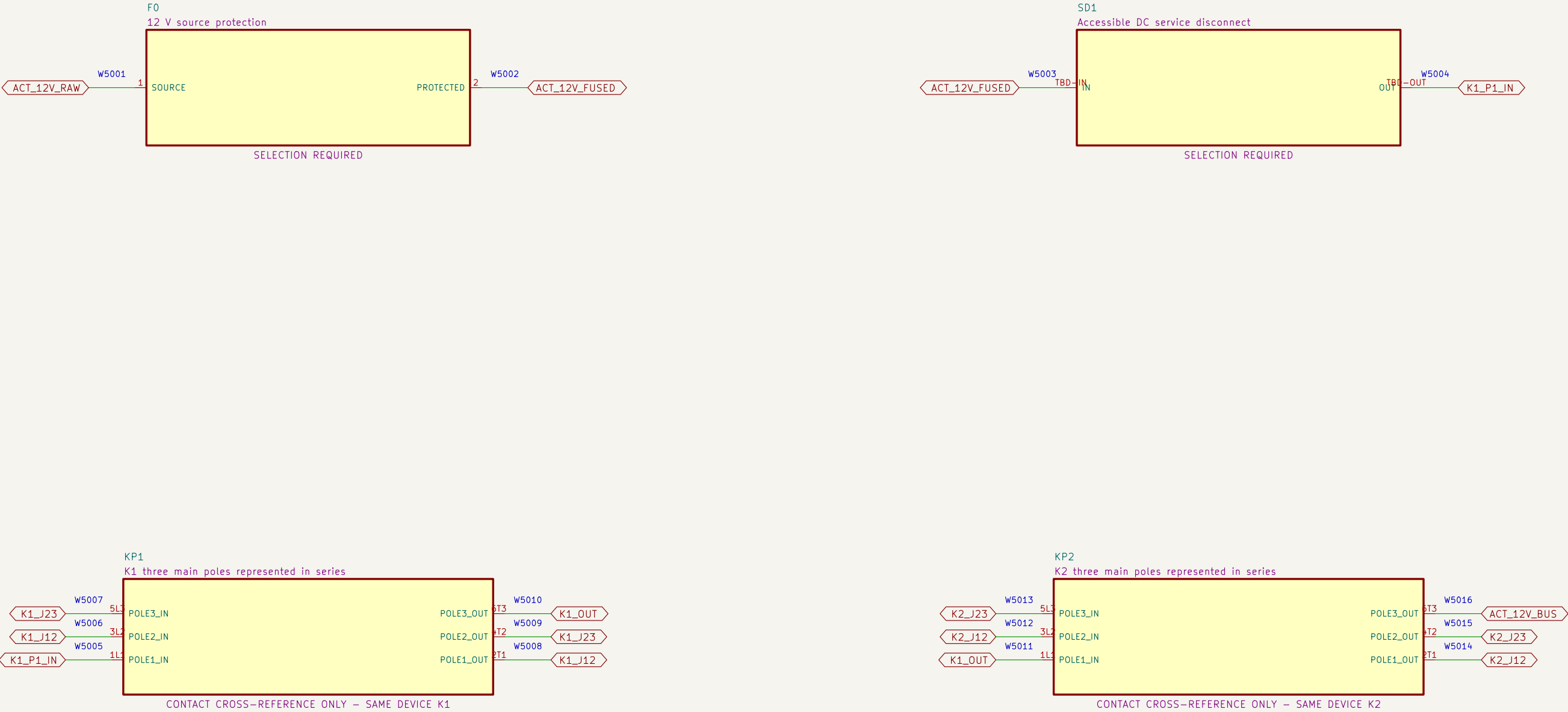
04 Contactor coils, mirror contacts and EDM

K1 and K2 are distinct final elements; their mirror contacts form the monitored restart return.



05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.

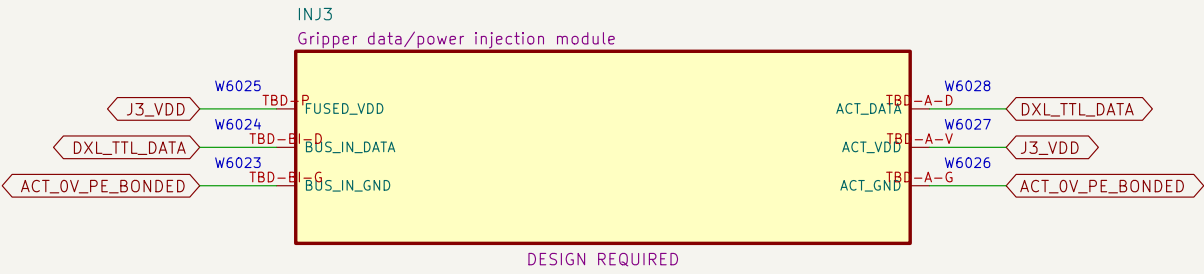
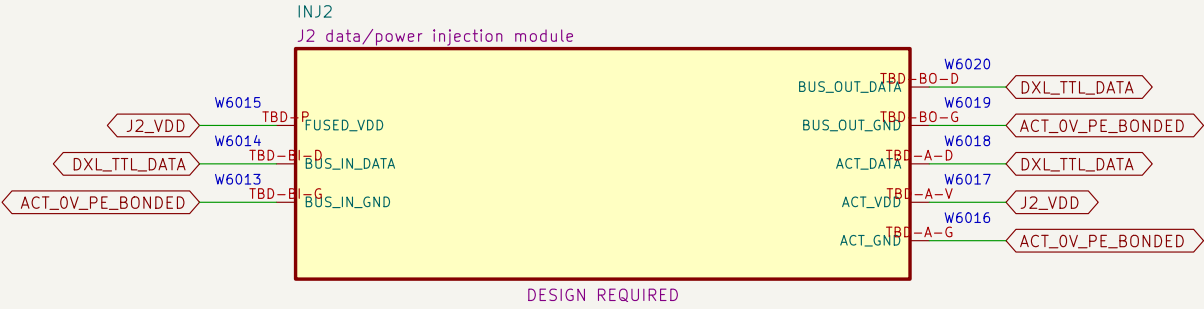
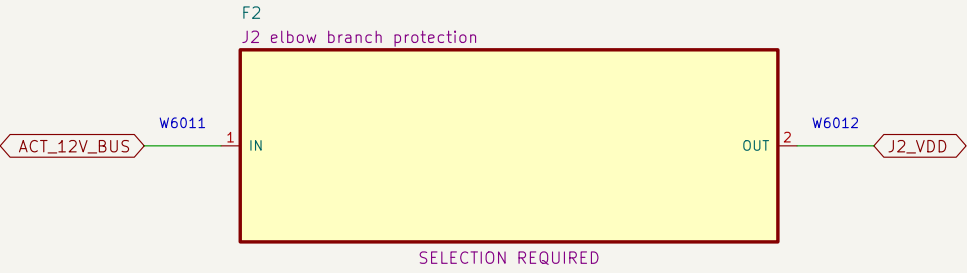
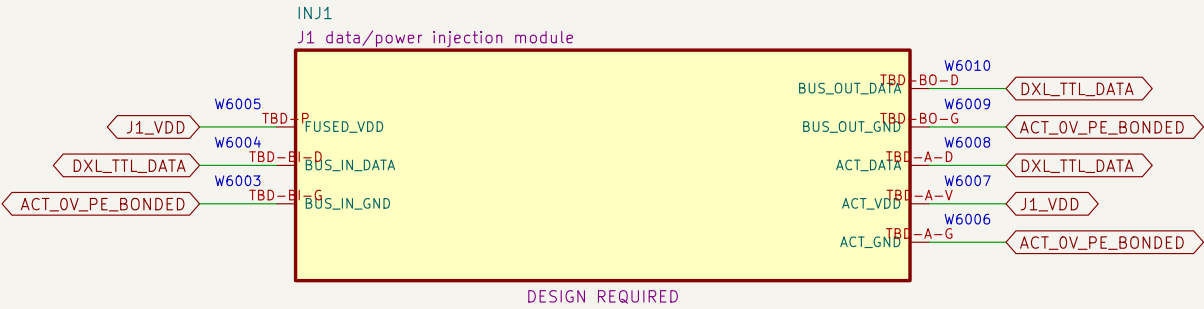
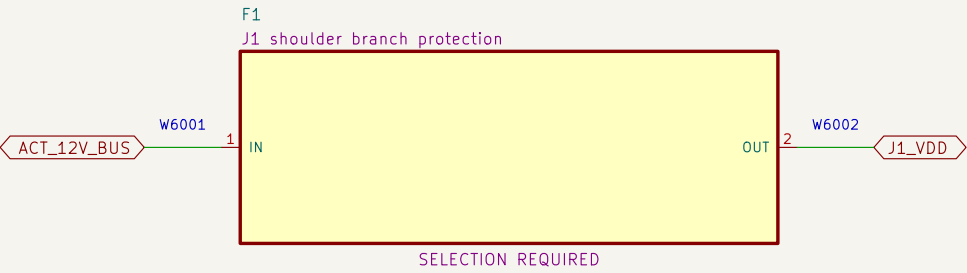


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|-----------------------------------------------------------------------------------------------|------------------|--------------|
| CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED                                        |                  |              |
| PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION                                    |                  |              |
| Project Button                                                                                |                  |              |
| Sheet: /05 Redundant actuator–power interruption/<br>File: 05_actuator_interruption.kicad_sch |                  |              |
| Title: PB HR–V0 Electrical V3 – 05                                                            |                  |              |
| Size: A3                                                                                      | Date: 2026–08–06 | Rev: V3–P0.6 |
| KiCad E.D.A. 10.0.5                                                                           |                  | Id: 6/11     |



06 Protected actuator branches and VDD-isolating injection

Each actuator has a separate protected VDD branch; inter-actuator data cables must omit VDD.



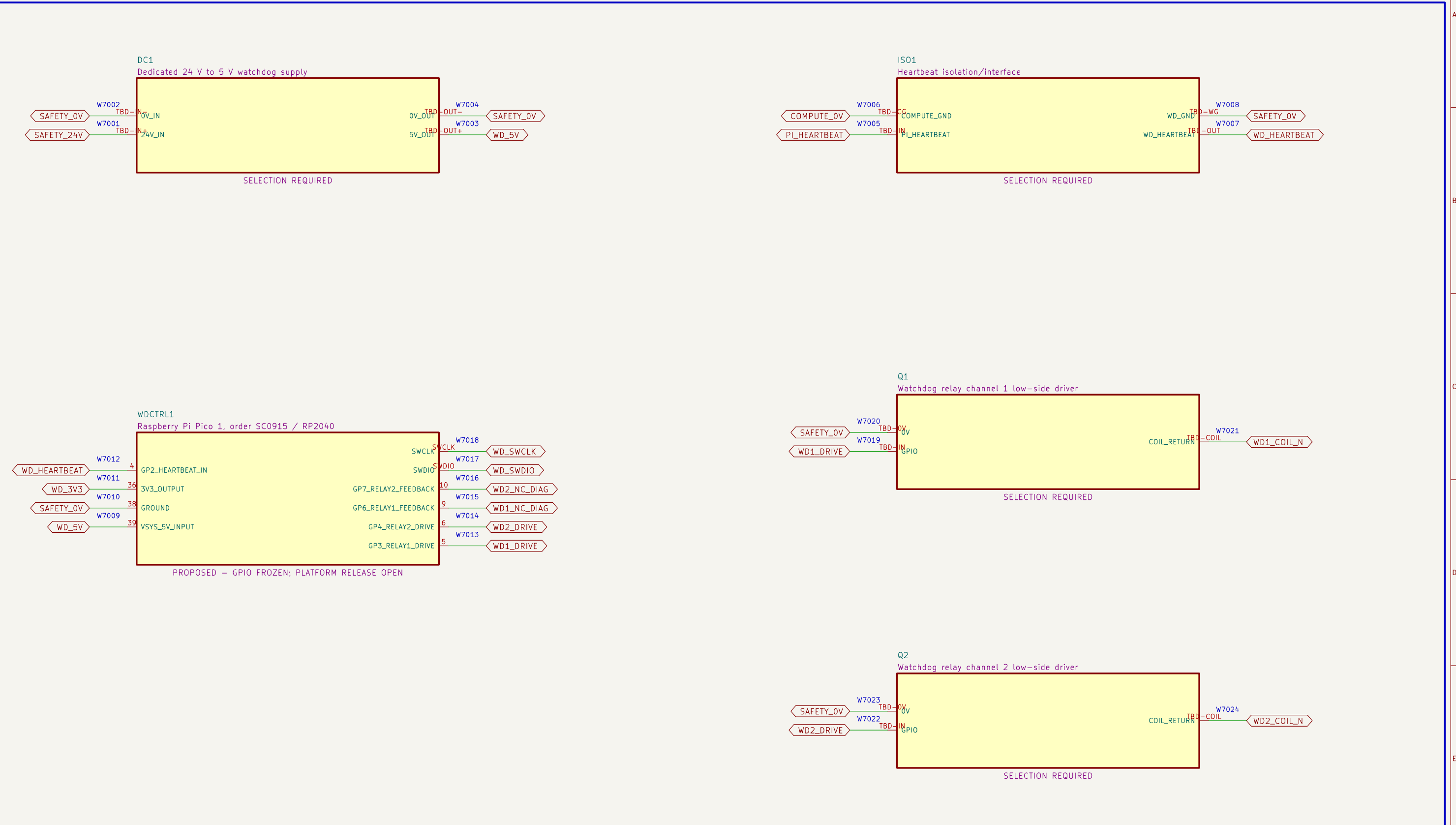
NOTE 1: Standard fully populated ROBOTIS TTL daisy-chain cables are prohibited because they would parallel branch VDD.

NOTE 2: Every injection module requires released source, continuity, isolation, pull and no-backfeed tests.

|                                                                                                                  |                  |              |
|------------------------------------------------------------------------------------------------------------------|------------------|--------------|
| CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED                                                           |                  |              |
| PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION                                                       |                  |              |
| Project Button                                                                                                   |                  |              |
| Sheet: /06 Protected actuator branches and VDD-isolating injection/<br>File: 06_branches_and_injection.kicad_sch |                  |              |
| Title: PB HR-V0 Electrical V3 – 06                                                                               |                  |              |
| Size: A3                                                                                                         | Date: 2026-08-06 | Rev: V3-P0.6 |
| KiCad E.D.A. 10.0.5                                                                                              |                  | Id: 7/11     |

07 Independent watchdog power, controller and drivers

The ordinary watchdog controller and drivers receive no safety–integrity credit by assertion.

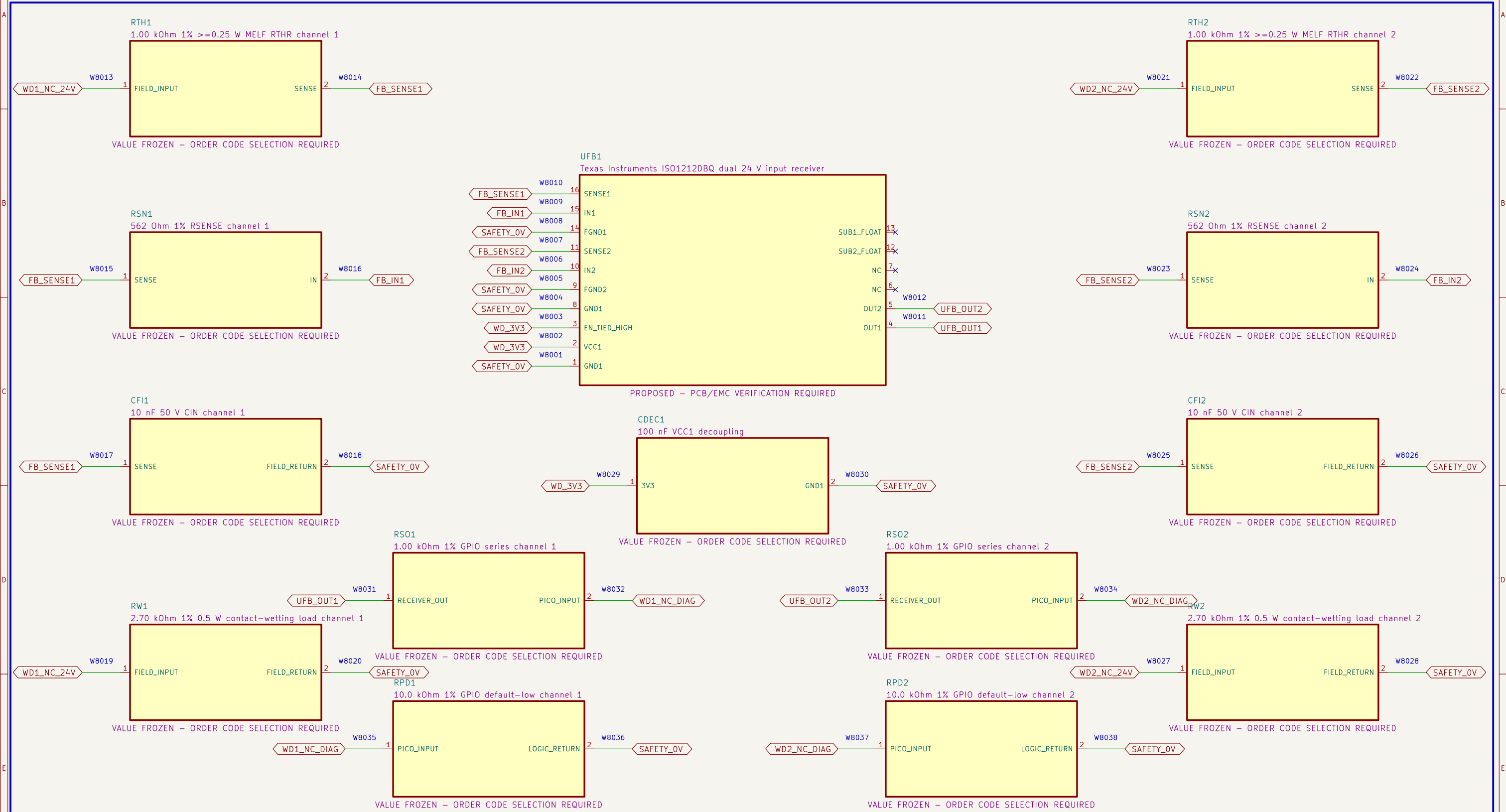


|                                                                                                       |                  |              |
|-------------------------------------------------------------------------------------------------------|------------------|--------------|
| CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED                                                |                  |              |
| PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION                                            |                  |              |
| Project Button                                                                                        |                  |              |
| Sheet: /07 Independent watchdog power, controller and drivers/<br>File: 07_watchdog_control.kicad_sch |                  |              |
| Title: PB HR–V0 Electrical V3 – 07                                                                    |                  |              |
| Size: A3                                                                                              | Date: 2026–08–06 | Rev: V3–P0.6 |
| KiCad E.D.A. 10.0.5                                                                                   | Id: 8/11         |              |



08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED  
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

**Project Button**

Sheet: /08 Calculated dual-channel 24 V watchdog feedback/  
File: 08\_watchdog\_feedback\_interface.kicad\_sch

**Title: PB HR-V0 Electrical V3 – 08**

Size: A3

Date: 2026-08-06

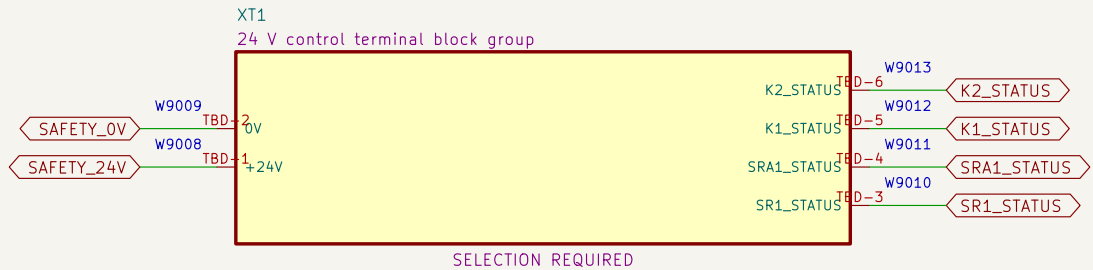
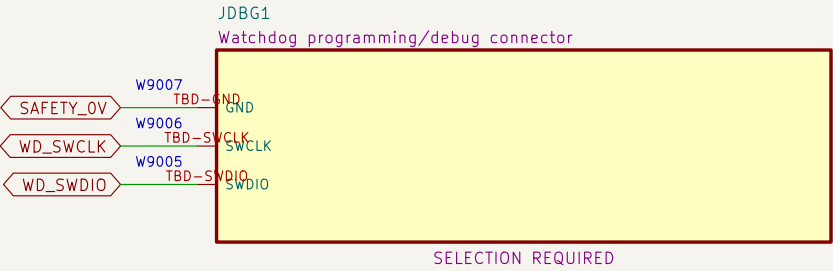
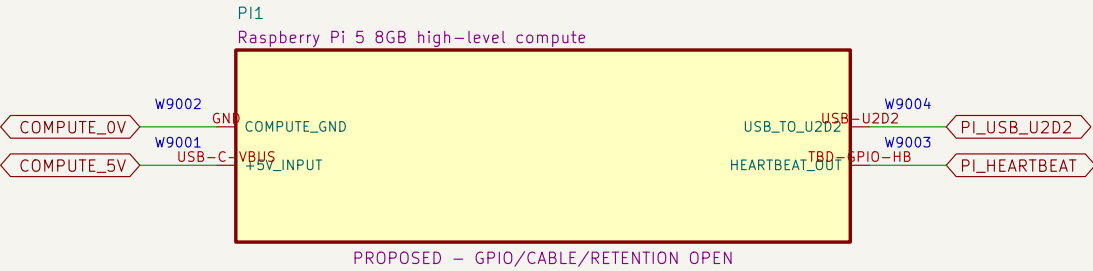
Rev: V3-P0.6

KiCad E.D.A. 10.0.5

Id: 9/11

09 Compute, debug and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



NOTE 1: Programming/debug connections must not enable outputs during operation.

NOTE 2: Terminal family, markers, conductor range, torque and enclosure layout remain selection required.

10 U2D2, actuator ports and bonding boundary

The U2D2 cable carries DATA and GND only; protected VDD is injected at each actuator.

